```
"**********************************************
"* DL Register Area - 120Word
"*
"* DW00000 - DW00009 (Pls Use)
"* DW00010 - DW00039 (Bit Use)
"* DW00040 - DW00089 (Word Use)
"* DW00090 - DW00119 (Timer)
"***********************************************

"# Program Nam     :  STR1 Cycle 10 (V.S/PreAlignment)
"# Functior        :
"# Call Prograr    :  MPS481
"# Sub Program     :
"# Func Program    :

VAR;
    LONG STR_L_CAM_MARK_I              %ML97300;
    LONG STR_R_CAM_MARK_I              %ML97330;

    BOOL STR_ALIGN_L_REQ               %MB970004;
    BOOL STR_ALIGN_R_REQ               %MB970005;

    BOOL STR_L_CAM_LINE_RE             %MB970024;
    BOOL STR_R_CAM_LINE_RE             %MB970025;

    BOOL STR_ALIGN_L_MANUAL_RE         %MB970044;
    BOOL STR_ALIGN_R_MANUAL_RE         %MB970045;

    BOOL STR_L_CAM_DL_ALIGN_RE         %MB970084;
    BOOL STR_R_CAM_DL_ALIGN_RE         %MB970085;

    BOOL STR_ALIGN_L_RESP              %MB970104;
    BOOL STR_ALIGN_R_RESP              %MB970105;

    BOOL STR_L_CAM_LINE_RE             %MB970124;
    BOOL STR_R_CAM_LINE_RE             %MB970125;

    BOOL STR_L_CAM_MANUAL_ALIGN_RI     %MB970144;
    BOOL STR_R_CAM_MANUAL_ALIGN_RI     %MB970145;

    BOOL STR_L_CAM_DL_ALIGN_RE         %MB970184;
    BOOL STR_R_CAM_DL_ALIGN_RE         %MB970185;

    BOOL STR_L_CAM_ALIGN_RESULT_(      %MB970600;
    BOOL STR_R_CAM_ALIGN_RESULT_(      %MB970700;

    BOOL STR_L_CAM_ALIGN_RESULT_N      %MB970601;
    BOOL STR_R_CAM_ALIGN_RESULT_N      %MB970701;

    LONG STR_L_CAM_ALIGN_ERROR_COI     %ML97062;      // 0:OK 1:NO MARK ID 2:MANUAL SEARCH 3:MANUAL OK 4:MANUAL ABORT
    LONG STR_R_CAM_ALIGN_ERROR_COI     %ML97072;      // 0:OK 1:NO MARK ID 2:MANUAL SEARCH 3:MANUAL OK 4:MANUAL ABORT

    BOOL STR_L_CAM_MANUAL_ALIGN_IN     %MB970622;
    BOOL STR_R_CAM_MANUAL_ALIGN_IN     %MB970722;

    BOOL STR_L_CAM_MANUAL_ALIGN_ABOI   %MB970624;
    BOOL STR_R_CAM_MANUAL_ALIGN_ABOI   %MB970724;

    LONG STR_L_CAM_ALIGN_SEARCH_TYI    %ML97314;      // 0:Mark Search 1:Edge Search 2:DL 3:Manual 4:Auto Mark Search
    LONG STR_R_CAM_ALIGN_SEARCH_TYI    %ML97344;

    BOOL STR_ALIGN_MANUALSTOP_ALARN    %MB075381;
    BOOL STR_ALIGN_MARK_DISTANCE_ALAI  %MB075382;
    BOOL STR_ALIGN_ZERO_VAL_ALARN      %MB075383;

    LONG STR_ALIGN_L_RESULT_XDAT       %ML97064;
    LONG STR_ALIGN_L_RESULT_YDAT       %ML97066;
    LONG STR_ALIGN_R_RESULT_XDAT       %ML97074;
    LONG STR_ALIGN_R_RESULT_YDATA      %ML97076;

END_VAR;
00000    BLK MW24160 MW97304 W8;
00001    BLK MW24160 MW97334 W8;

00002    CLR DW00050 W20;
00003    ML07892 = 0;                  " PreAlign Auto Mark Search X Offset Result
00004    ML07894 = 0;                  " PreAlign Auto Mark Search Y Offset Result

00005    STR_ALIGN_L_REQ = 0           " Alignment Request OFF(Left)
00006    STR_ALIGN_R_REQ = 0           " Alignment Request OFF(Right)
00007    STR_L_CAM_LINE_REQ = 0;
00008    STR_R_CAM_LINE_REQ = 0;
```

```
00009       STR_ALIGN_L_MANUAL_REQ = 0;
00010       STR_ALIGN_R_MANUAL_REQ = 0;
00011       STR_L_CAM_DL_ALIGN_REQ = 0;
00012       STR_R_CAM_DL_ALIGN_REQ = 0;

00013       STR_L_CAM_MARK_ID = GL51018;
00014       STR_R_CAM_MARK_ID = GL51018;

00015       IF !MB076323 == 1                                          " PreAlignment Po
00016           MSEE MPS480;
00017           TIM TMW30430;
00018       IEND;


00019       IOW STR_ALIGN_L_RESP | STR_ALIGN_R_RESP == 0;

        //  IF !MB000800 == 1                                          " No-Load Operatio
00020           DB00020 = 0;
00021           DB00021 = 0;
00022           DW00020 = 0;

00023           WHILE DW00020 < MW30440;
00024               STR_L_CAM_ALIGN_SEARCH_TYPE = 4;
00025               STR_R_CAM_ALIGN_SEARCH_TYPE = 4;

00026               STR_ALIGN_L_REQ = 1                                " Alignment Request ON(Left

00027               STR_ALIGN_R_REQ = 1                                " Alignment Request ON(Right)
00028               IOW STR_L_CAM_ALIGN_ERROR_CODE == 0;
00029               IOW STR_R_CAM_ALIGN_ERROR_CODE == 0;

00030               IOW STR_ALIGN_L_RESP & STR_ALIGN_R_RESP==          " Alignment Respons O

00031               IF STR_L_CAM_ALIGN_RESULT_OK == 1
00032                   ML07810 = STR_ALIGN_L_RESULT_XDATA             " Alignment Data X1
00033                   ML07812 = STR_ALIGN_L_RESULT_YDATA * -         " Alignment Data Y1
00034                   DB00020 = 1;
00035               IEND;
00036               IF STR_R_CAM_ALIGN_RESULT_OK == 1;
00037                   ML07814 = STR_ALIGN_R_RESULT_XDATA             " Alignment Data X2
00038                   ML07816 = STR_ALIGN_R_RESULT_YDATA * -         " Alignment Data Y2
00039                   DB00021 = 1;
00040               IEND;

00041               MB07860A = STR_L_CAM_ALIGN_RESULT_NG;
00042               MB07860B = STR_L_CAM_ALIGN_RESULT_N

00043               STR_ALIGN_L_REQ = 0                                " Alignment Request OFF(Left)

00044               STR_ALIGN_R_REQ = 0                                " Alignment Request OFF(Right)


00045               IF DB00020 & DB00021 == 1;
00046                   DW00020 = MW30440 + 1;
00047                   IF CW00054 == 1;
00048                       MB07860A = 0;
00049                       MB07860B = 0;
00050                   IEND;
00051               ELSE;
00052                   DW00020 = DW00020 + 1;
00053                   TIM T100;
00054               IEND;
00055           WEND;
00056           IF GL51020 == 0                                        " Auto Mark Search (0:Use 1:Unuse)
00057               IF (MB07860A | MB07860B)== 1;
00058                   IF MB078606 == 0                               " Auto Mark Seach Complete
00059                       MSEE MPS485;
00060                   IEND;
00061               IEND;
00062           IEND;

                " Manual Search Flag
00063           IF MB07860A | MB07860B == 1;

00064               DB000000 = 0;
00065               DB000010 = 0;
00066               DB000011 = 0;
00067               WHILE DB000000 == 0
00068                   STR_L_CAM_ALIGN_SEARCH_TYPE = 0;
00069                   STR_R_CAM_ALIGN_SEARCH_TYPE = 0;

00070                   IOW STR_ALIGN_L_RESP | STR_ALIGN_R_RESP ==         " Wait Alignment Respons OFF

00071                   STR_ALIGN_L_REQ = MB07860A                         " L Camera Manual Alignment Request
00072                   STR_ALIGN_R_REQ = MB07860B                         " R Camera Manual Alignment
```

```
Request
                EOX;

                IOW (STR_L_CAM_ALIGN_RESULT_OK | STR_L_CAM_MANUAL_ALIGN_ING) & (STR_R_CAM_ALIGN_RESU
LT_OK | STR_R_CAM_MANUAL_ALIGN_ING) ==           " Alignment Respons O

                IF STR_L_CAM_ALIGN_RESULT_OK & STR_R_CAM_ALIGN_RESULT_OK == 1;

                ELSE;
                    PFORK Line1 Line2;
                    Line1:
                        IF STR_L_CAM_MANUAL_ALIGN_ING == 1;
                            IOW !STR_L_CAM_MANUAL_ALIGN_ING == 1;
                            IOW STR_ALIGN_L_RESP == 1;
                        IEND;
                        JOINTO LabelX1;

                    Line2:
                        IF STR_R_CAM_MANUAL_ALIGN_ING == 1;
                            IOW !STR_R_CAM_MANUAL_ALIGN_ING == 1;
                            IOW STR_ALIGN_R_RESP == 1;
                        IEND;
                        JOINTO LabelX1;
                    LabelX1: PJOINT;

                    IF STR_L_CAM_MANUAL_ALIGN_ABORT == 1;
                        MB000015 = 1;                                    " Cycle Stop
                        STR_ALIGN_MANUALSTOP_ALARM = 1                   " Alignment Resp NG
                        STR_ALIGN_L_REQ = 0;
                        DB000000 = 1;
                        EOX;
                    IEND;
                    IF STR_R_CAM_MANUAL_ALIGN_ABORT == 1;
                        MB000015 = 1;                                    " Cycle Stop
                        STR_ALIGN_MANUALSTOP_ALARM = 1                   " Alignment Resp NG
                        STR_ALIGN_R_REQ = 0;
                        DB000000 = 1;
                        EOX;
                    IEND;
                IEND;

                IF STR_L_CAM_ALIGN_RESULT_OK == 1;
                    ML07810 = STR_ALIGN_L_RESULT_XDATA + (IL8C16 - ML6480                " Alignm
ent Data X1
                    ML07812 = STR_ALIGN_L_RESULT_YDATA * -1 - (IL8816 - ML6322           " Alignm
ent Data Y1
                    MB07860A = 0;                                                        " L Alig
nment NG (0:OK 1:NG)
                    STR_ALIGN_L_MANUAL_REQ = C                                           " L Came
ra Manual Alignment Request
                IEND;

                IF STR_R_CAM_ALIGN_RESULT_OK == 1;
                    ML07814 = STR_ALIGN_R_RESULT_XDATA + (IL8C96 - ML6500                " Alignm
ent Data X2
                    ML07816 = STR_ALIGN_R_RESULT_YDATA * -1 - (IL8816 - ML6322           " Alignm
ent Data Y2
                    MB07860B = 0;                                                        " R Alig
nment NG (0:OK 1:NG)
                    STR_ALIGN_R_MANUAL_REQ = C                                           " R Came
ra Manual Alignment Request
                IEND;

                STR_ALIGN_L_REQ = 0                                                      " L Came
ra Manual Alignment Request OFF
                STR_ALIGN_R_REQ = 0                                                      " R Came
ra Manual Alignment Request OFF
                IOW STR_ALIGN_L_RESP | STR_ALIGN_R_RESP ==                               " Wait A
lignment Respons OFF

                IF MB07860A | MB07860B == 0;
                    DB000000 = 1;
                IEND;
            EOX;
            WEND;
        IEND;
        IF !MB000015 == 1                                                " Cycle Stop
            IF ML07810 == 0                                              " Alignment ZeroValue Check
                IF ML07812 == 0;
                    IF ML07814 == 0                                      " Alignment ZeroValue Check
                        IF ML07816 == 0;
                            STR_ALIGN_ZERO_VAL_ALARM = 1                 " Alignment ZeroValue Alarm
                            EOX;
                        IEND;
                    IEND;
                IEND;
            IEND;
```

```

00136              DF00050 = IL8C16 + ML0781C                                    " Left Mark Real X Po

00137              DF00052 = IL8C96 + ML07814;                                   " Right Makr Real X Pos
00138              DF00054 = ML07812 - ML0781(                                   " L,R Mark Y Distance
00139              DL00056 = GL51016                                             " Align Mark Pos D

00140              DF00058 = DF00052 - DF00050;

00141              DF00060 = (DF00058 * DF00058) + (DF00054 * DF00054);
00142              DF00062 = SQT(DF00060);
00143              DL00064 = DL00056 - DF00062;

00144              IF DL00064 < 0;                                               " Find Mark Distance Check
00145                  DL00064 = -DL00064;
00146              IEND;

00147              IF DL00064 > ML30434;
00148                  STR_ALIGN_MARK_DISTANCE_ALARM = 1;
00149              IEND;

00150          IEND;

00151      MSEE MPS465;                                                      " Stick Transfer 1 Axis Settir
      //  ELSE;
      //      ML07810 = 0;                                                   " Alignment Data X1
      //      ML07812 = 0;                                                   " Alignment Data Y1
      //      ML07814 = 0;                                                   " Alignment Data X2
      //      ML07816 = 0;                                                   " Alignment Data Y
      //  IEND;

00152  RET;
```